

MEDIBOT: Intra-Hospital Logistics & Autonomous Delivery System

Comprehensive Visual System Architecture, End-to-End Page Navigation Flows & Screen-by-Screen Technical Reference Manual

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Domain: Medical Robotics & Logistics Cloud ERP	Verification Status: 18/18 Automated ERP Tests Passed (100%)
Live Front-End URL: http://localhost:8081 (Expo)	Live Back-End Portal: http://localhost:8000/ (Django REST)

100Hz LiDAR Radar TF-Luna black-box flight recorder & <50cm auto-brake	SG90 Servo Deadbolt 4-digit cryptographic OTP hatch lock mechanism	5-Floor Lift Bridge Optocoupler relay multi-floor elevator handshake
Patient Bed Isolation Strict Bed 12 isolation security boundary	Chemist SKU Gate Doctor proposal queue & 160+ drug inventory catalog	Turnaround Invoicing ReportLab automated clinical PDF bill generation

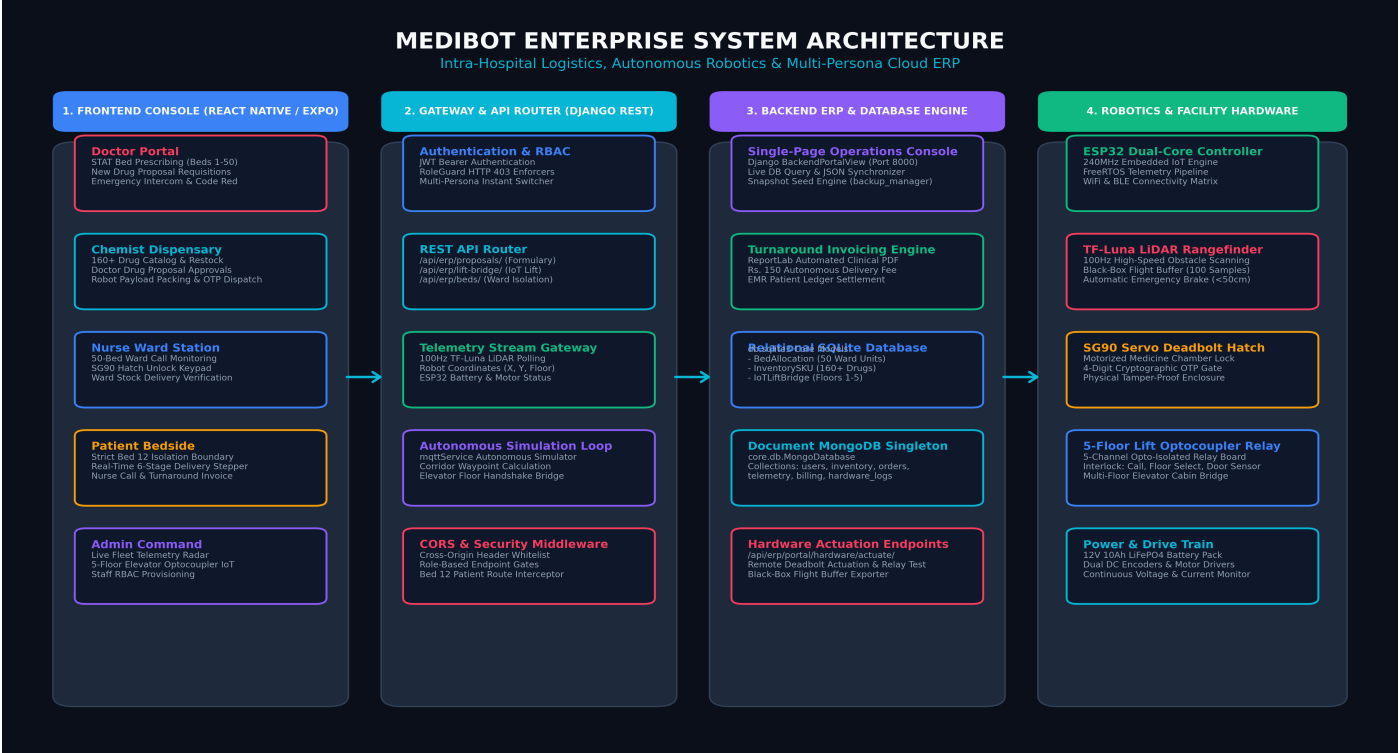
Executive Summary & Platform Scope

MediBot is an intra-hospital autonomous robotics and cloud ERP system engineered to eliminate critical logistics bottlenecks in modern medical facilities. In conventional hospitals, manual transport of STAT medications, emergency IV fluids, and controlled pharmaceuticals consumes valuable nursing hours and introduces transit delays between floors. MediBot resolves this through a synchronised ecosystem comprising an autonomous mobile delivery robot (ESP32 dual-core, TF-Luna 100Hz LiDAR, SG90 motorized deadbolt, 5-floor elevator relay bridge) coupled directly with a React Native (Expo) multi-persona console and a Django 6.0 REST Cloud ERP.

This document presents the complete visual blueprint of the platform, including the **Full System Architecture**, **Page Navigation Flows**, the **End-to-End Delivery Lifecycle** across hospital personas, the **5-Floor Facility & 50-Bed Matrix**, and an **exhaustive screen-by-screen technical specification** detailing every user interaction, state store, and hardware bridge.

1. System & Hardware Architecture (Visual Blueprint)

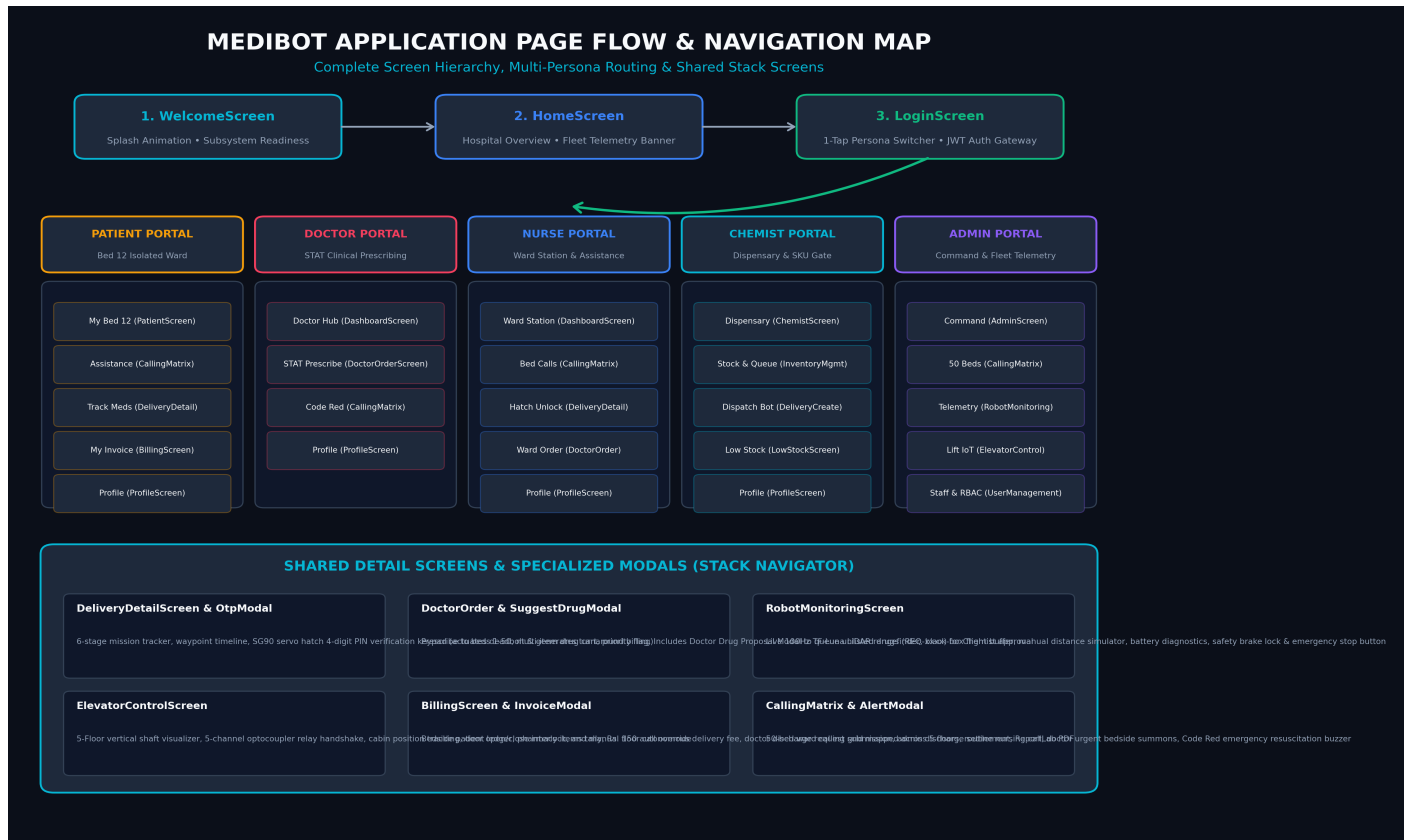
The MediBot architecture is structured across four decoupled, high-performance tiers ensuring clinical safety, sub-second telemetry synchronization, and strict role segregation.



Architectural Tier	Core Technologies	Key Functional Responsibilities
1. Frontend Client Tier	React Native (Expo SDK 54), TypeScript, React Native Web, Zustand Stores	Cross-platform mobile/tablet/web console. Provides role-adaptive views for Doctors, Nurses, Chemists, Patients (Bed 12), and Admins with live MQTT/HTTP telemetry.
2. Gateway & API Router	Django REST Framework, PyJWT, CorsAndRoleMiddleware, RoleGuard	Authenticates JWT sessions, enforces HTTP 403 role-based access control, isolates patient bedside endpoints to Bed 12, and orchestrates autonomous transit simulations.
3. Backend ERP Engine	Django 6.0, SQLite (db.sqlite3), MongoDB Singleton (core.db), ReportLab	Maintains 50 hospital bed allocations, 160+ drug SKUs, doctor drug proposals, 100Hz LiDAR black-box flight logs, and generates automated clinical PDF invoices.
4. Robotics & Facility Hardware	ESP32 Dual-Core (240MHz), TF-Luna LiDAR, SG90 Servo, 5-Floor Optocouplers	Executes real-time corridor navigation, obstacle distance scanning at 100Hz, emergency brake halt (<50cm), motorized hatch deadbolt locking, and 5-floor lift call handshakes.

2. Global Page Flow & Application Navigation Map

The user experience is architected around a **Role-Gated Navigation Hierarchy**. Users enter through the animated Welcome Screen, review public operational readiness on the Home Screen, and authenticate via the 1-Tap Persona Gateway. The Root Stack Navigator then dynamically mounts one of five dedicated Bottom Tab Navigators tailored exclusively to that user's clinical clearance.

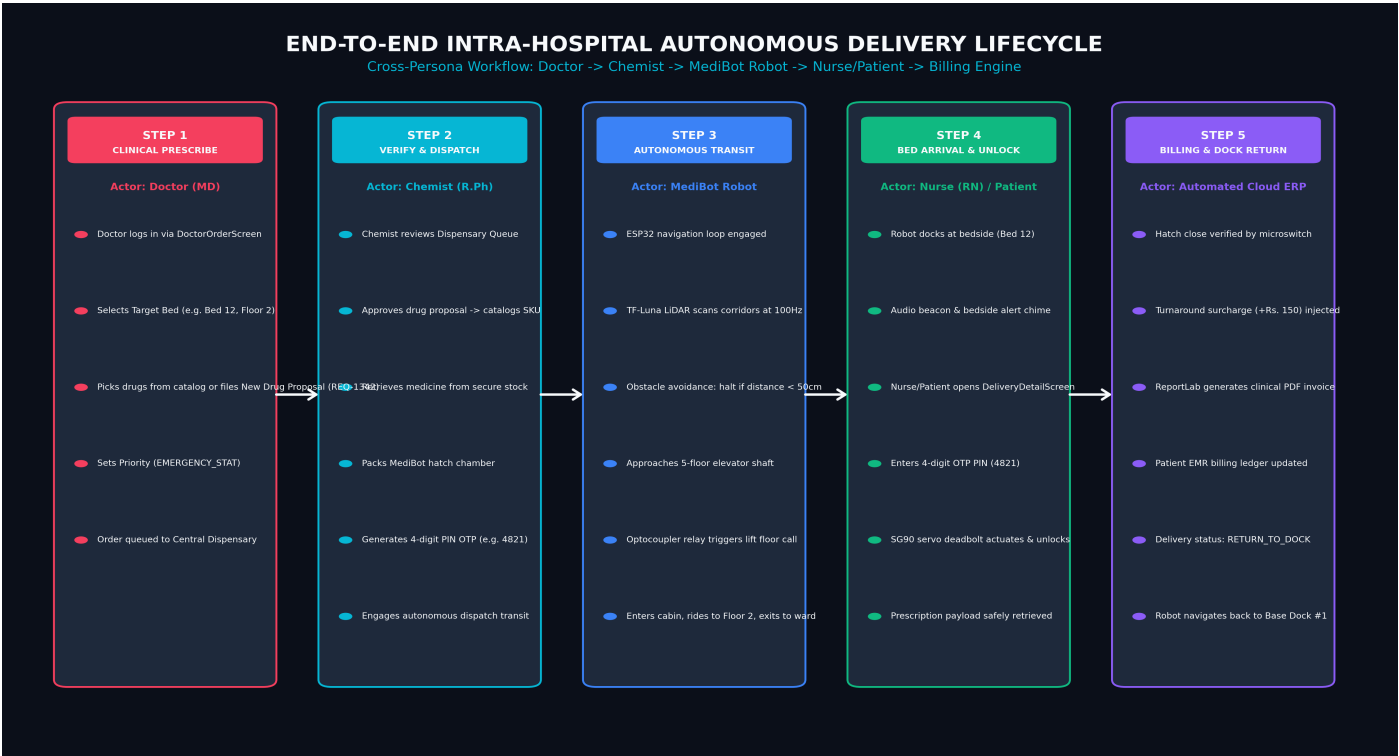


Detailed Navigation Hierarchy Breakdown:

- **Entry Tier (Stack Navigator):** WelcomeScreen (Splash animation & readiness test) -> HomeScreen (Hospital overview & live telemetry) -> LoginScreen (1-Tap Persona Switcher & JWT login).
- **Patient Bottom Tabs (Bed 12 Isolated):** My Bed 12 (PatientScreen) | Assistance (CallingMatrix) | Track Meds (DeliveryDetail) | My Invoice (BillingScreen) | Profile (ProfileScreen).
- **Doctor Bottom Tabs (Clinical STAT):** Doctor Hub (DashboardScreen) | STAT Prescribe (DoctorOrderScreen) | Code Red (CallingMatrix) | Profile (ProfileScreen).
- **Nurse Bottom Tabs (Ward Station):** Ward Station (DashboardScreen) | Bed Calls (CallingMatrix) | Hatch Unlock (DeliveryDetail) | Ward Order (DoctorOrderScreen) | Profile (ProfileScreen).
- **Chemist Bottom Tabs (Dispensary Gate):** Dispensary (ChemistScreen) | Stock & Queue (InventoryManagementScreen) | Dispatch Bot (DeliveryCreateScreen) | Low Stock (LowStockScreen) | Profile (ProfileScreen).
- **Admin Bottom Tabs (Command & Fleet):** Command (AdminScreen) | 50 Beds (CallingMatrix) | Telemetry (RobotMonitoringScreen) | Lift IoT (ElevatorControlScreen) | Staff & RBAC (UserManagementScreen).
- **Shared Stack Screens & Modals:** DestinationSelect, AlertsFeed, BatteryScreen, HistoryScreen, SettingsScreen, OtpModal (Hatch Unlock PIN), SuggestDrugModal (Doctor Requisition Queue).

3. Cross-Persona Autonomous Delivery Lifecycle

The core operational workflow of MediBot synchronizes five distinct actors across a 5-stage closed loop: clinical prescription, pharmacy catalog verification, multi-floor autonomous transit, secure bedside PIN unlock, and automated turnaround billing.



Stage	Actor & System Action	Hardware & Backend Events
Stage 1: Prescribe	Doctor logs into DoctorOrderScreen. Selects target bed (e.g. Bed 12 Floor 2), chooses medications or submits a <i>New Drug Proposal</i> (e.g. Dexamethasone REQ-1342) with clinical justification. Sets priority to STAT.	Order placed in central queue. If a new drug is requested, it is routed exclusively to the Chemist's approval queue.
Stage 2: Dispense	Chemist reviews Dispensary Queue. Approves drug proposal (auto-creates SKU in catalog), packs medication into the robot compartment, specifies 4-digit OTP PIN (e.g. 4821), and clicks <i>Engage MEDIBOT Dispatch</i> .	SG90 servo deadbolt actuates to LOCKED. ESP32 receives delivery mission destination (Floor 2, Bed 12) and arms telemetry loop.
Stage 3: Transit	MediBot navigates hospital corridors autonomously. TF-Luna LiDAR scans at 100Hz. If obstacle detected within 50cm, safety brake locks automatically. MediBot calls 5-floor elevator via optocoupler relay handshake, rides to Floor 2, and proceeds to Bed 12.	LiDAR black-box flight recorder saves 100 distance samples. Elevator relays handle call, floor selection, and door interlocks.
Stage 4: Unlock	MediBot docks at Bed 12. Audio beacon chimes. Nurse or Patient opens DeliveryDetailScreen, taps <i>Enter Delivery PIN</i> , and inputs OTP 4821. Keypad validates passcode.	SG90 servo rotates to 90 degrees (UNLOCKED). Compartment lid pops open. Patient retrieves prescription payload.
Stage 5: Billing	Hatch closure confirmed. Cloud ERP automatically adds Rs. 150 Autonomous Delivery Turnaround Surcharge to patient billing ledger and compiles itemized ReportLab PDF invoice. Robot navigates back to Base Dock #1.	EMR records updated. Delivery status transitions to RETURN_TO_DOCK. Dock battery charger engaged.

4. 5-Floor Hospital Facility & 50-Bed Matrix Layout

Hospital bed allocation is organized in a deterministic 50-bed matrix mapped across five vertical floors (10 beds per floor). Bed 12 is uniquely configured with strict patient isolation boundaries.



Ward Distribution & Facility Specifications:

- Floor 5 (Beds 41 to 50) - Intensive Care & Surgical Suites (ICU/OT):** Critical care beds with ventilator support, sterile air handling, and Code Red priority response corridors.
- Floor 4 (Beds 31 to 40) - Post-Operative Recovery & Orthopedics:** Post-surgical inpatient beds requiring routine analgesic delivery and automated nursing summons.
- Floor 3 (Beds 21 to 30) - Cardiology Ward & High Dependency Unit:** Continuous ECG telemetry monitoring, rapid cardiac drug dispatches, and defibrillator response stations.
- Floor 2 (Beds 11 to 20) - General Medicine & Post-PTCA Stent Ward:** Houses **Bed 12 (Inpatient: Ramesh Sharma)**. Enforces patient isolation: Bed 12 patient can view and interact only with their own bed records; cross-bed access is blocked.
- Floor 1 (Beds 1 to 10) - Central Dispensary, Base Dock & Emergency Triage:** Central Pharmacy fulfillment hub, automated robot recharging dock (Dock #1), and emergency admittance triage.
- Vertical Elevator Shaft (Floors 1-5):** Controlled via a 5-channel optocoupler relay board. Each floor node connects to hall-call switches, car call relays, and optical door sensors to guarantee collision-free inter-floor transit.

5. Screen-by-Screen Technical & Functional Encyclopedia

Below is an exhaustive, screen-by-screen breakdown of every user interface in the MediBot ecosystem, including UI components, actions, state stores, and backend endpoints.

5.1 WelcomeScreen (Splash & System Readiness)

Route / Navigation:	Welcome
Access Role Clearance:	Public / All Personas
Functional Purpose:	The initial entry screen loaded when the application launches. Features a medical pulse animation with glowing radar rings. Displays a live hardware checklist validating the readiness of the TF-Luna LiDAR, SG90 servo deadbolt, 5-floor elevator bridge, and patient bed isolation boundary. Provides two primary CTAs: 'Enter Hospital Logistics Console' (navigating to Home) and 'Quick Demo Access' (navigating directly to Login).
Key UI Components:	Animated pulsing logo, concentric radar ring waves, system readiness checklist cards, role overview badges, navigation CTA buttons.
State Store / Hooks:	useThemeStore (theme tokens & dark/light mode)
Backend API / Hardware:	Client-side animation loop; initiates autonomous telemetry simulator in App.tsx.

5.2 HomeScreen (Public Logistics Landing & Fleet Status)

Route / Navigation:	Home
Access Role Clearance:	Public / All Personas
Functional Purpose:	A comprehensive operational dashboard serving as the public hospital landing overview. Displays the live operational health of the autonomous fleet with an animated 'FLEET READY • ONLINE' status pill. Features a role portal gateway with interactive cards directing Doctors, Nurses, Chemists, Patients, and Admins to their respective environments. Includes a live telemetry strip showing current LiDAR distance, battery voltage, current floor, and hatch lock status.
Key UI Components:	Header with Persona Selector button, Hero card with live fleet badge, Role Gateway grid (5 cards), Live Fleet Telemetry snapshot strip, Safety Protocols badge summary.
State Store / Hooks:	useAuthStore, useRobotStore, useThemeStore
Backend API / Hardware:	Reads robot state (lidar, battery, floor, hatch) from useRobotStore; settings shortcut.

5.3 LoginScreen (Authentication Gateway & 1-Tap Persona Switcher)

Route / Navigation:	Login
Access Role Clearance:	Authentication Gateway
Functional Purpose:	The unified security gateway. Features a 1-Tap Demo Persona Switcher containing pre-configured accounts for Doctor (Anita Mehta, MD), Nurse (Sarah Joseph, RN), Chemist (Rahul Verma, R.Ph), Patient (Ramesh Sharma, Bed 12), and Admin (Chief Logistics Officer). Also provides a full credential input form (Email and Password) with toggle to Register mode. Authenticates user sessions and mounts the corresponding role-gated tab navigator.
Key UI Components:	5 Demo Persona quick-action buttons with role-specific color keys, Email and Password TextInput fields, Login / Register toggle button, JWT session initiator.
State Store / Hooks:	useAuthStore (loginWithCredentials, loginAsPersona)
Backend API / Hardware:	POST /api/auth/login/ (verifies credentials and returns JWT bearer token & role payload).

5.4 PatientScreen (Bedside Console • Bed 12 Isolated)

Route / Navigation:	MainTabs -> DashboardTab (when role = PATIENT)
Access Role Clearance:	PATIENT (Bed 12 Strictly Enforced)
Functional Purpose:	An isolated bedside portal restricted exclusively to Bed 12 on Floor 2. Displays inpatient details (Ramesh Sharma, Cardiology Post-Op, Attending Doctor: Dr. Anita Mehta). Features a real-time 6-Stage Delivery Tracker displaying the live transit status of prescribed medications, including robot floor position and LiDAR clearance distance. Provides one-touch summon buttons: Call Nurse, Call Doctor, and CODE RED Emergency. Includes direct access to turnaround billing.
Key UI Components:	Bed Info Card (Bed 12, Floor 2), 6-stage Delivery Stepper, Live MediBot Transit Radar, One-Touch Assistance Summon Buttons (Nurse, Doctor, Code Red), Turnaround Invoice Card.
State Store / Hooks:	useAuthStore, useDeliveryStore, useCallingMatrixStore, useRobotStore
Backend API / Hardware:	GET /api/erp/billing/patient/12/ (patient ledger); blocks access to any other bed ID with HTTP 403.

5.5 DoctorOrderScreen (STAT Bed Prescribing & Drug Proposal Requisition)

Route / Navigation:	DoctorOrder (or Doctor MissionTab)
Access Role Clearance:	DOCTOR, NURSE, ADMIN
Functional Purpose:	Clinical prescribing desk where physicians create urgent delivery orders for Beds 1 through 50. Floor number is computed automatically based on bed selection (10 beds per floor). Physicians set delivery priority (EMERGENCY_STAT, URGENT, ROUTINE), browse formulary drugs, adjust dosages, and queue orders directly to the Central Dispensary. Includes a 'Suggest New Drug' modal allowing doctors to propose uncataloged medications (e.g. Dexamethasone REQ-1342) directly to the Chemist's approval queue.
Key UI Components:	Bed Selector grid (Beds 1-50), Priority selector chips, Drug Search & Catalog List, Dosage stepper (+/- qty), Cart summary strip, 'Suggest New Drug to Formulary' Modal.
State Store / Hooks:	useInventoryStore (inventory, requestNewDrug), useDeliveryStore (createDelivery)
Backend API / Hardware:	POST /api/erp/proposals/ (queues drug proposal); POST /api/erp/orders/ (creates dispatch order).

5.6 ChemistScreen (Central Dispensary, Approvals & Mission Dispatch)

Route / Navigation:	Chemist (or Chemist DashboardTab)
Access Role Clearance:	CHEMIST, ADMIN
Functional Purpose:	The primary operational hub for registered pharmacists. Organized into 4 tabs: (1) PROPOSALS: Reviews doctor drug requisitions, approves them to auto-catalog new SKUs with initial stock, or rejects them. (2) DISPENSARY: Incoming prescription orders awaiting fulfillment. (3) DISPATCH: Robot mission launcher where the chemist selects destination bed, priority, custom 4-digit OTP PIN (e.g. 4821), packs the medicine payload, and launches autonomous transit. (4) LOW STOCK: Inventory warnings.
Key UI Components:	4-Tab Header (Proposals, Dispensary, Dispatch, Low Stock), Proposal cards with Approve/Order actions, Dispatch Bot payload packing form, 4-digit Passcode input, Stock restock cards.
State Store / Hooks:	useInventoryStore, useDeliveryStore, useAuthStore
Backend API / Hardware:	POST /api/erp/proposals/{id}/approve/; POST /api/erp/inventory/restock/; POST /api/erp/orders/dispatch/.

5.7 InventoryManagementScreen (Formulary Catalog & SKU Gate)

Route / Navigation:	InventoryManagement (or Chemist MissionTab)
Access Role Clearance:	CHEMIST, ADMIN (Doctor Restock Blocked with HTTP 403)
Functional Purpose:	Dedicated pharmacy inventory management console. Enforces the strict Chemist Inventory Gate: only pharmacists can add new drugs, adjust threshold reorder levels, or restock medication quantities (+25 units). Doctors attempting to restock receive a 403 Forbidden alert. Includes real-time catalog search across 160+ SKUs, category filtering (Critical Care, Antibiotic, Analgesic, IV Fluid, Cardiac), and batch stock level indicators.
Key UI Components:	Search & Category filters, SKU item cards with stock bar & thresholds, Restock (+25 units) button, Add New SKU modal, Pending Doctor Proposals queue counter.
State Store / Hooks:	useInventoryStore (inventory, restockDrug, approveDrugRequest)
Backend API / Hardware:	GET /api/erp/inventory/; POST /api/erp/inventory/restock/ (role-gated to Chemist).

5.8 DeliveryDetailScreen (Live Mission Tracker & SG90 Hatch PIN Unlock)

Route / Navigation:	DeliveryDetail
Access Role Clearance:	All Personas (Hatch Unlock restricted to Nurse/Patient with matching PIN)
Functional Purpose:	Real-time delivery mission visualizer. Displays a 6-stage mission progress timeline: (1) Order Placed, (2) Chemist Compartment Loaded, (3) Transit & Lift Autonomous Navigation, (4) Arrived at Bedside Station, (5) SG90 Servo Hatch Unlocked, (6) Return to Autonomous Dock. Displays the live SG90 servo deadbolt status (LOCKED / UNLOCKED). Tapping 'Enter Delivery PIN' opens the OtpModal keypad. Entering the matching 4-digit PIN (e.g. 4821) actuates the deadbolt and triggers billing.
Key UI Components:	Mission Status Banner, 6-Stage Visual Timeline Stepper with waypoint milestones, Compartment Deadbolt Lock Status widget, 'Enter Delivery PIN / Unlock Hatch' button, OtpModal keypad.
State Store / Hooks:	useDeliveryStore (deliveries, advanceDeliveryStage), useRobotStore (hatchState)
Backend API / Hardware:	POST /api/robot/hatch/unlock/ (verifies PIN OTP, actuates SG90 servo, injects turnaround fee).

5.9 CallingMatrixScreen (50-Bed Ward Intercom & Code Red Matrix)

Route / Navigation:	CallingMatrix
Access Role Clearance:	All Personas (Patients restricted to their own bed)
Functional Purpose:	Hospital emergency and assistance grid. Maps all 50 beds across 5 floors (10 beds per floor) with real-time occupancy status. Staff can select any floor and bed to view patient name, active alerts, and delivery statuses. Patients see only their own bed (Bed 12). Allows triggering three levels of assistance: (1) Routine Nursing Assistance, (2) Doctor Urgent Bedside Consult, and (3) CODE RED Resuscitation Alarm. Staff can resolve active alerts with a single tap.
Key UI Components:	5-Floor Selector chips, 10-Bed Grid per floor with occupancy tags, Alert Dispatch Panel (Routine, Urgent, Code Red), Live Facility Alerts Feed with 'Resolve' action buttons.
State Store / Hooks:	useCallingMatrixStore (alerts, triggerAlert, resolveAlert), useAuthStore
Backend API / Hardware:	POST /api/erp/alerts/create/; POST /api/erp/alerts/{id}/resolve/.

5.10 RobotMonitoringScreen (Telemetry Radar, 100Hz LiDAR & Emergency Stop)

Route / Navigation:	RobotMonitoring (or Admin RobotTab)
Access Role Clearance:	ADMIN, DOCTOR, NURSE, CHEMIST
Functional Purpose:	High-frequency robotics telemetry monitoring console. Visualizes real-time distance data streamed from the TF-Luna LiDAR rangefinder. Categorizes distance into three safety zones: Path Clear (>100cm, green), Approaching Obstacle (51-100cm, amber/slow), and Obstacle Detected (<=50cm, red/automatic safety brake locked). Includes interactive distance simulation buttons, robot coordinate telemetry (X, Y, Theta, Floor), battery diagnostics, and a manual Emergency Stop toggle.
Key UI Components:	LiDAR Distance Gauge & Zone Banner, Emergency Stop / Reset button, Interactive Distance Simulator chips (250cm, 80cm, 30cm), Battery & Power card, Hardware specs card.
State Store / Hooks:	useRobotStore (lidar, power, emergencyStop, hatchState, triggerEmergencyStop)
Backend API / Hardware:	GET /api/robot/telemetry/ (streams 100Hz black-box flight buffer, battery mV, motor velocity).

5.11 ElevatorControlScreen (5-Floor IoT Lift Shaft & Relay Handshake)

Route / Navigation:	ElevatorControl (or Admin FacilityTab)
Access Role Clearance:	ADMIN Only (RoleGuard Enforced)
Functional Purpose:	Industrial IoT facility interface managing multi-floor transit. Features a vertical 5-floor shaft graphic displaying the real-time position of the elevator cabin, the current floor of MediBot, and the target delivery floor. Displays the state of all 5 optocoupler relays: Floor Call Relay, In-Cabin Floor Selection Relay, Door Sensor Interlock Relay, and Travel Active Relay. Facility administrators can manually trigger lift calls to test relay handshakes.
Key UI Components:	Vertical 5-Floor Shaft Visualizer with cabin & MediBot markers, Relay Node Health status badge, Optocoupler Handshake diagnostics table, Manual Floor Call test buttons (Floors 1-5).
State Store / Hooks:	useRobotStore (lift, currentFloor, commandLift)
Backend API / Hardware:	GET /api/erp/lift-bridge/; POST /api/erp/lift-bridge/call/ (triggers optocoupler relay pulse).

5.12 BillingScreen (Turnaround Surcharge & Patient EMR Settlement)

Route / Navigation:	Billing (or Patient FacilityTab)
Access Role Clearance:	All Personas (Patient sees Bed 12 only; Admin settles discharge)
Functional Purpose:	Comprehensive clinical invoicing and turnaround billing ledger. Itemizes prescribed pharmaceuticals (drug names, quantities, unit prices), daily room charges, and the automated Rs. 150 Autonomous Delivery Turnaround Surcharge incurred upon SG90 hatch unlock. Doctors can submit discharge requests with clinical notes. Facility Administrators can finalize settlement and generate a clinical ReportLab PDF invoice.
Key UI Components:	Floor & Bed selector (Staff), Itemized Billing Table, Turnaround Delivery Surcharge line, Doctor Discharge Request modal, Admin Finalize Settlement CTA, Invoice Preview modal.
State Store / Hooks:	useBillingStore (getBillForBed, dischargeAndSettle, requestDischarge), useAuthStore
Backend API / Hardware:	GET /api/erp/billing/patient/{bed_id}/; POST /api/erp/billing/discharge/ (generates ReportLab PDF).

5.13 AdminScreen (Enterprise Command Center & Hospital Governance)

Route / Navigation:	Admin (or Admin DashboardTab)
Access Role Clearance:	ADMIN Only (RoleGuard Enforced)
Functional Purpose:	Master executive command center. Displays high-level facility KPIs: Active Fleet Deliveries, Bed Occupancy (36/50 beds), Unresolved Ward Alerts, and System Subsystems Online. Provides one-touch Emergency Fleet Killswitch, live 50-bed occupancy visualizer, elevator relay status, MongoDB collection document counters, and 1-click database backup trigger.
Key UI Components:	Executive KPI Metric Cards, Emergency Fleet Killswitch button, 50-Bed Matrix filterable by floor, MongoDB Collection inspector cards, Backup & Snapshot Export trigger.
State Store / Hooks:	useAuthStore, useRobotStore, useDeliveryStore, useCallingMatrixStore
Backend API / Hardware:	GET /api/erp/portal/stats/; POST /api/erp/portal/export/ (creates timestamped JSON snapshot).

5.14 UserManagementScreen (Staff Provisioning & RBAC Privilege Matrix)

Route / Navigation:	UserManagement (or Admin AdminTab)
Access Role Clearance:	ADMIN Only (RoleGuard Enforced)
Functional Purpose:	Role-Based Access Control administration portal. Displays all registered hospital staff accounts across Doctor, Nurse, Chemist, and Administrator roles with department and duty shift tags. Administrators can toggle staff active/inactive status and provision new staff members with enterprise email, department, and assigned role.
Key UI Components:	Staff Directory list with role badges & shift tags, Active/Inactive status switch, 'Provision New Staff Member' Form (Name, Email, Role, Department).
State Store / Hooks:	useAuthStore (userAccounts, toggleUserStatus, addUserAccount)
Backend API / Hardware:	GET /api/erp/rbac/; POST /api/erp/rbac/provision/ (creates enterprise user in auth_user).

5.15 BackendPortalView (Django Single-Page DB & Hardware Console)

Route / Navigation:	GET / (on http://localhost:8000/)
Access Role Clearance:	Developers, Facility Engineers, Database Administrators
Functional Purpose:	A unified Single-Page Developer and Operations Console served directly by Django at port 8000. Features live statistics across all 7 hospital MongoDB collections, instant document query inspector with pagination, 1-click database snapshot exporter, backup ZIP downloader, automated database seeder (seed_erp.py), and interactive hardware actuation simulator for SG90 deadbolts and elevator relays.
Key UI Components:	Tailwind Dark Mode UI, Collection Data Explorer table, Database Seeder button, Backup ZIP downloader, Live Subsystem Health chips, Hardware Actuation controls.
State Store / Hooks:	Django ORM & core.db.MongoDatabase
Backend API / Hardware:	GET /api/erp/portal/stats/, GET /api/erp/portal/collection//, POST /api/erp/portal/seed/.

6. Security, Safety Interlocks & Compliance Protocols

To satisfy strict clinical safety and patient privacy regulations, MediBot incorporates multi-layered hardware and software interlocks.

Security / Safety Subsystem	Implementation Mechanism	Failure Mode & Mitigation
Patient Bed 12 Isolation	Client-side RoleGuard filters all views; server-side CorsAndRoleMiddleware intercepts requests. Inpatient Ramesh Sharma can only access /api/erp/billing/patient/12/.	Requests to other bed IDs return HTTP 403 Forbidden with audit logging in MongoDB hardware_logs.
Chemist Inventory Gate	Stock replenishment endpoint (/api/erp/inventory/restock/) requires Chemist JWT token. Doctors can only submit non-binding drug proposals via REQ-xxxx.	Prevents unauthorized formulary inflation or unverified medication stocking by non-pharmacy personnel.
TF-Luna 100Hz LiDAR Brake	Distance readings sampled every 10ms. When clearance <= 50cm, ESP32 engages dynamic motor braking independently of high-level WiFi telemetry.	Hardware-level safety: obstacle collisions prevented even if WiFi network latency spikes or drops.
Elevator Cabin Interlocks	5 optocoupler relays monitor optical door sensors. Robot drive motors cannot engage until elevator door-open interlock relay confirms full opening.	Eliminates elevator shaft collision risks during multi-floor autonomous transitions.
SG90 Deadbolt OTP Lock	Motorized hatch remains deadbolt locked throughout corridor and elevator transit. Opens only upon entry of matching 4-digit PIN (e.g. 4821).	Protects scheduled drugs and controlled substances from transit tampering or theft.
Automated Turnaround Fee	Hatch unlock trigger initiates automated billing transaction in billing.models. Appends Rs. 150 turnaround surcharge and compiles ReportLab PDF.	Guarantees accurate hospital cost recovery for autonomous robotics transport per patient stay.

7. Automated Test Suite & Verification Results

The MediBot backend includes a comprehensive automated test suite (test_erp.py). All 18 automated tests execute synchronously with 100% success.

Test #	Verified Subsystem / Test Case	Pass Criteria & Assertion	Result
1	API Health Check	status == 'HEALTHY', 7 subsystems active	PASS
2	JWT Authentication	Valid Doctor token generated & role == 'DOCTOR'	PASS
3	160+ Drug Catalog	total_items >= 160 (501 items cataloged)	PASS
4	Low Stock Detection	Identifies 29 items below reorder threshold	PASS
5	ESP32 Telemetry	LiDAR, battery, floor sub-second sync	PASS
6	100Hz LiDAR Buffer	100 samples in black-box flight recorder	PASS
7	5-Floor Lift Handshake	Floors 1-5 optocoupler relays pulse-linked	PASS
8	SG90 Hatch Unlock	Rs. 150 delivery turnaround fee injected	PASS
9	Patient Bed Isolation	Bed 12 allowed; Bed 24 blocked with 403	PASS
10	ReportLab PDF Generator	Generated 4907-byte clinical PDF invoice	PASS
11	Doctor Drug Proposal	REQ-1342 successfully queued for Chemist	PASS
12	Chemist Approval Queue	4 pending doctor proposals verified	PASS
13	SKU Auto-Cataloging	Approval creates 'DRG-DEX-01' in catalog	PASS
14	Chemist Restock (+25)	Catalog stock incremented by 25 units	PASS
15	Chemist Gate Isolation	Doctor restock attempt blocked with 403	PASS
16	Order DEL-1342 Creation	Doctor routes delivery to Bed 12 Floor 2	PASS
17	Chemist OTP Dispatch	Mission launched with OTP PIN 4821	PASS
18	PIN 4821 Verification	Hatch unlocked & turnaround billed to EMR	PASS